Chuang Yang

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✧ Contact

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✧ Personal Details

Name: Chuang Yang (Family name: Yang, Given name: Chuang)

Language: Mandarin Chinese, English (College English Test Band 6, CET-6)

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✧ Academic Details

- **Major**: Navigation, Guidance and Control
- **Professional association**: Member of Royal Institute of Navigation, Member of Chinese Society of Aeronautics and Astronautics, Member of Chinese Association for Artificial Intelligence
- **Research interests:**
  Brain-inspired navigation & swarm navigation for robotics, Neuromorphic swarm, Inertial navigation system(INS), Integrated navigation system (INS/GNSS/Magnetic)
- **Current Ph.D. thesis topic: Brain-inspired swarm navigation for UAV Swarm**

Abstract: This research project (www.neuromorphicswarm.com) will fuse neural basis of bat's 3D navigation and swarm navigation and the conventional integrated navigation by employing neuroscience-oriented artificial intelligence method to develop a new class of brain-inspired swarm navigation architecture—towards Neuromorphic Swarm—of UAV Swarm.

✧ Skill Profile

- **Software and Programming Language**
  Matlab, C, LabVIEW, Python
- **Open Source Program**
  RatSLAM: an open source brain-based SLAM system
  NeuroSLAM: a brain-inspired SLAM system for 3D environments
- **Hardware**
  Arduino, STM32
- **Design of electronic and electrical system**
  (*I became interested in electronic products in childhood, greatly influenced by my parents who were all maintanance workers for electronic products, such as television, amplifier, etc.*)
  Two seclected designs of electronic system:

1. Third Prize in the 4th Petroleum Equipment Innovative Design Competition for China postgraduate (**team leader**), 09/2017, as shown in Fig.1.
2. Second Prize in the 12th Electronic Design Competition for China postgraduate (**team leader**), 07/2017, as shown in Fig.2.

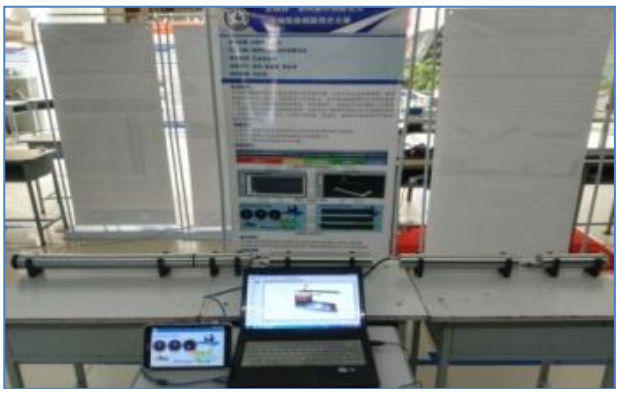
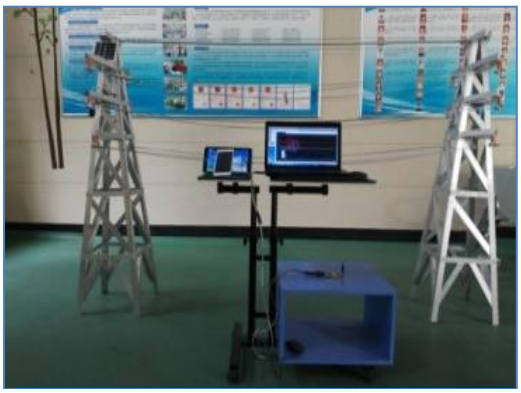

Fig.1(Left): **INS/Magnetic-based integrated navigation system of measurement while drilling (**LabVIEW+Arduino+sensors of IMU/Magnetic**)**

Fig.2(Right): **INS-based on-line monitoring system of transmission line galloping (**LabVIEW+Arduino+sensors of IMU**)**

✧ Education Experiences

- **04/2018—present**
  Nanjing University of Aeronautics and Astronautics, China
  PhD student of Navigation, Guidance and Control
  Supervisor: Prof. Jianye Liu
  Thesis Topic: Research on brain-inspired swarm navigation for multi-robot
- **09/2015—12/2017**
  Henan Polytechnic University, China
  Master of Navigation, Guidance and Control (***I received my M.S. degree ahead of schedule by six months due to the excellent academic performance***)
  Supervisor: Dr. Jinxian Yang
  Final thesis: Research on INS/magnetic integrated navigation technique for measurement while drilling
- **09/2011—07/2015**
  Henan Institute of Science and Technology, China
  Bachelor of Electrical Engineering and Automation (GPA: 3.68/4.0)

✧ Seclected Project Experiences

- **Regarding brain-inspired navigation & swarm navigation**

1. China, National Natural Science Foundation of China, Grant No. 61873125. **Brain-inspired intelligent cooperative navigation for dense UAV swarm,** 01/2019-12/2022. (main participant, my Ph.D. thesis topic).
2. China, Shanghai Aerospace Science and Technology Innovation Fund, Grant No. SAST2019-085. **Brain-inspired positioning technology for intelligent aircraft**, 12/2019-12/2021. (main participant).
3. China, National Natural Science Foundation of China, Grant No. 61973160. **Collaborative navigation optimization method of swarm-UAVs based on target-driven approaches in GNSS-denied environment**, 01/2020-12/2023. (participant)

- **Regarding integrated navigation**

4. China, National Natural Science Foundation of China, Grant No. 61973160. **Key Technologies of fault tolerant navigation based on dynamic performance evaluation for UAV swarm**, 01/2018-12/2020.

(participant)

5. China, Key Science and Technology Program of Henan Province, Grant No. 172102210289. **Research on on-line monitoring system of transmission line galloping based on inertial technology**, 09/2016-09/2017. (main participant)
6. China, National Natural Science Foundation of China, Grant No. U1404510. **Key technologies of MEMS inertial navigation for measurement while drilling**, 01/2015-12/2017. (main participant)

✧ Seclected Publications

- **Brain-inspired navigation & swarm navigation**

1. **Chuang Yang**, Zhi Xiong, Jianye Liu, Lijun Chao, Yudi Chen. Brain-inspired metric navigation of UAV based on model of multi-scale grid cells[J]. IEEE Transactions on Cognitive and Developmental Systems (Under review, 2020.08)
2. **Chuang Yang**, Jianye Liu, Zhi Xiong, Lijun Chao. Brain-inspired vector navigation method for UAV based on model of multi-scale grid cells[J]. Journal of Chinese Inertial Technology, 2020, 28(02):179-185. (EI, in Chinese)
3. **Chuang Yang**, Jianye Liu, Zhi Xiong. A new paradigm of brain-inspired navigation integrating perception and decision[EB/OL]. article.xuexi.cn, 2020.03.04. (in Chinese)
4. **Chuang Yang**, Jianye Liu, Zhi Xiong, Jun Xiong. Brain-inspired navigation technology integrating perception and action decision: A review and outlook[J]. Acta Aeronautica et Astronautica Sinica, 2020, 41(1):323280. doi:10.7527/S1000-6893.2019.23280. (EI, in Chinese)
5. Jianye Liu, **Chuang Yang**, Zhi Xiong, Jizhou Lai, Jun Xiong. Attractor neural network-based brain-inspired navigation technology for UAV[J]. Navigation Positioning & Timing, 2019, 6(5): 52-60. (in Chinese)
6. **Chuang Yang**, Jianye Liu. Brain-inspired navigation technology and its applications in unmanned autonomous system[J]. Naas & Inertal Technology, 2019, 200+201:151-152. (in Chinese)

- **Integrated navigation**

7. Jinxian Yang, **Chuang Yang**, Shuanglei Li, et al. Vibration analysis of transmission lines based on MIMU[J]. Journal of Vibration and Shock, 2018, 37(11): 230-236. (EI, in Chinese)
8. Jinxian Yang, **Chuang Yang**, Jinbao Jiang. A robust mag/INS-based orientation estimation algorithm for measurement while drilling[J]. IEEE Sensors Journal, 2017, 17(23): 7971–7980. (SCI)
9. Jinxian Yang, **Chuang Yang**. MEMS gyro signal reconstruction via layerwise and adaptive MP[J]. Optics and Precision Engineering, 2017, 25(12): 3160-3168. (EI, in Chinese)
10. Jinxian Yang, **Chuang Yang**, Zhitao Jiang. Adaptive extraction of gravity information for drilling tool with magnetic and inertial integrated system[J]. Journal of china coal society, 2017, 42(12): 3331-3337. (EI, in Chinese)
11. Jinxian Yang, **Chuang Yang**. Signal extraction of mems gyro utilized in MWD based on sparse representation[J]. Chinese Journal of Sensors & Actuators, 2017, 30(11): 1677-1683. (in Chinese)
12. Jinxian Yang, **Chuang Yang**. Gyro drift estimation and compensation based on gravity quaternion[J]. Chinese Journal of Sensors and Actuators, 2017, 30(8): 1187-1192. (in Chinese)

✧ Honors and Awards

10/2017: National Scholarship for Postgraduate

09/2017: Third Prize in the 4th Petroleum Equipment Innovative Design Competition for China postgraduate, *INS/Magnetic-based integrated navigation system of measurement while drilling*. (**team leader**)

07/2017: Second Prize in the 12th Electronic Design Competition for China postgraduate, *INS-based on-line monitoring system of transmission line galloping*. (**team leader**)

09/2015-01/2018: Full scholarship for Postgraduate

11/2013: National Scholarship for Encouragement

03/2013: Third-level Certificate for National Computer

11/2012: National Scholarship for Encouragement

09/2011-07/2015: The First Prize Scholarship for four years